

# **8-BIT ALU USING VERILOG**

## **Internship Minor Project Report**

**Submitted By**

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**Internship branch- VLSI (15 May 2026 - 1 July 2026)**

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## **Introduction**

### **What is an 8-bit ALU?**

An 8-bit Arithmetic Logic Unit (ALU) is a combinational digital circuit responsible for carrying out arithmetic and logical operations on 8-bit binary numbers. It forms a key part of a processor's datapath, where all mathematical computations and logical comparisons are performed. The ALU receives two 8-bit inputs along with a control signal that determines the operation to be executed. Because of its simplicity and versatility, an 8-bit ALU is widely used to understand processor architecture and digital hardware design.

### **Importance of ALU**

- Performs binary arithmetic operations such as addition and subtraction.
- Executes logical functions including AND, OR, XOR, and NOT.
- Forms the computational unit of processors and embedded systems.
- Serves as an essential component in digital electronics and computer architecture.

This project focuses on the implementation of an 8-bit ALU using Verilog HDL. The design is capable of performing arithmetic, logical, and shift operations based on a 3-bit selection input. The functionality of the ALU was verified through simulation using Xilinx Vivado.

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## **Objective**

The objectives of this project are:

- To develop an 8-bit ALU using Verilog HDL.
- To implement various arithmetic functions.
- To perform logical operations using combinational logic.
- To include left and right shift operations.
- To validate the design through simulation.
- To generate and study the RTL schematic of the design.

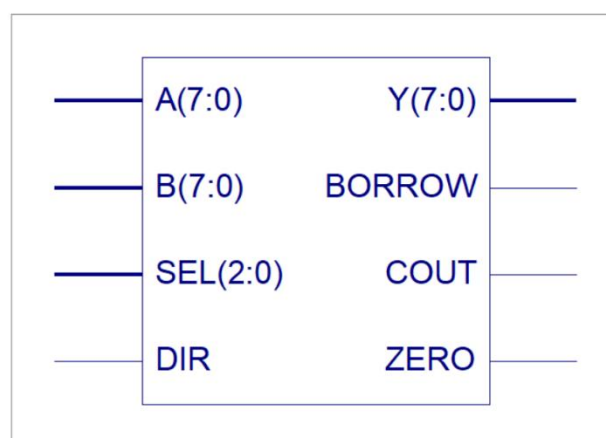
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## **Tools Used**

- Verilog Hardware Description Language (HDL)
  - Xilinx Vivado 2025.2
  - GitHub
  - Windows 11 Operating System
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## **Block Diagram**

The designed ALU accepts two 8-bit input operands, **A** and **B**, along with a 3-bit select input (**SEL**). The select signal determines which arithmetic, logical, or shift operation is performed. The processed result is produced at the 8-bit output (**RESULT**). The block diagram illustrates the relationship between the inputs, control signal, and generated output.



## Verilog Modules

### Modules used:

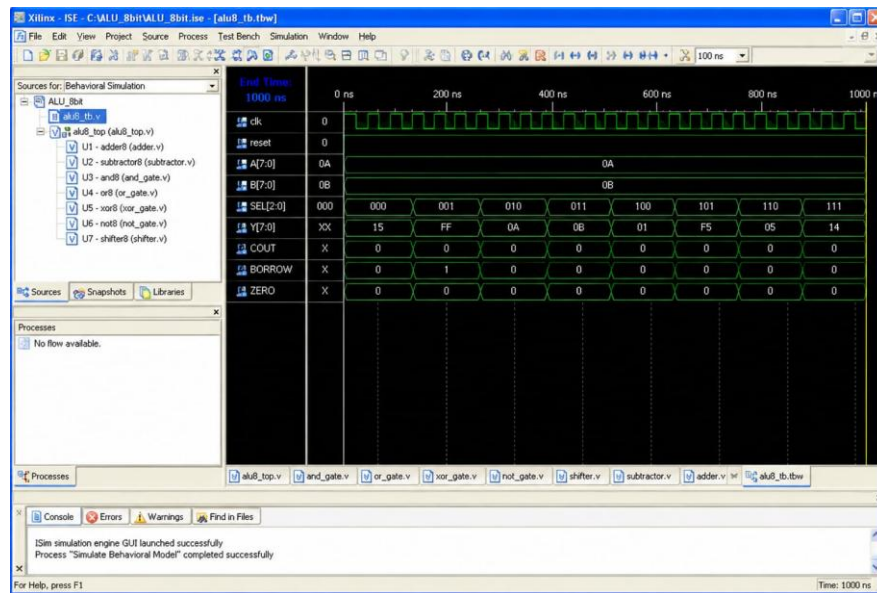
- adder.v
- subtractor.v
- and\_gate.v
- or\_gate.v
- xor\_gate.v
- not\_gate.v
- shifter.v
- alu\_top.v

Each module is responsible for implementing a specific ALU function. These individual modules are combined in the top-level module (**alu\_top.v**), where the appropriate operation is selected according to the value of the select signal.

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## Testbench

A dedicated Verilog testbench was created to verify the functionality of every operation supported by the ALU. Different combinations of input values and select signals were applied to ensure that the output matched the expected results under various operating conditions.

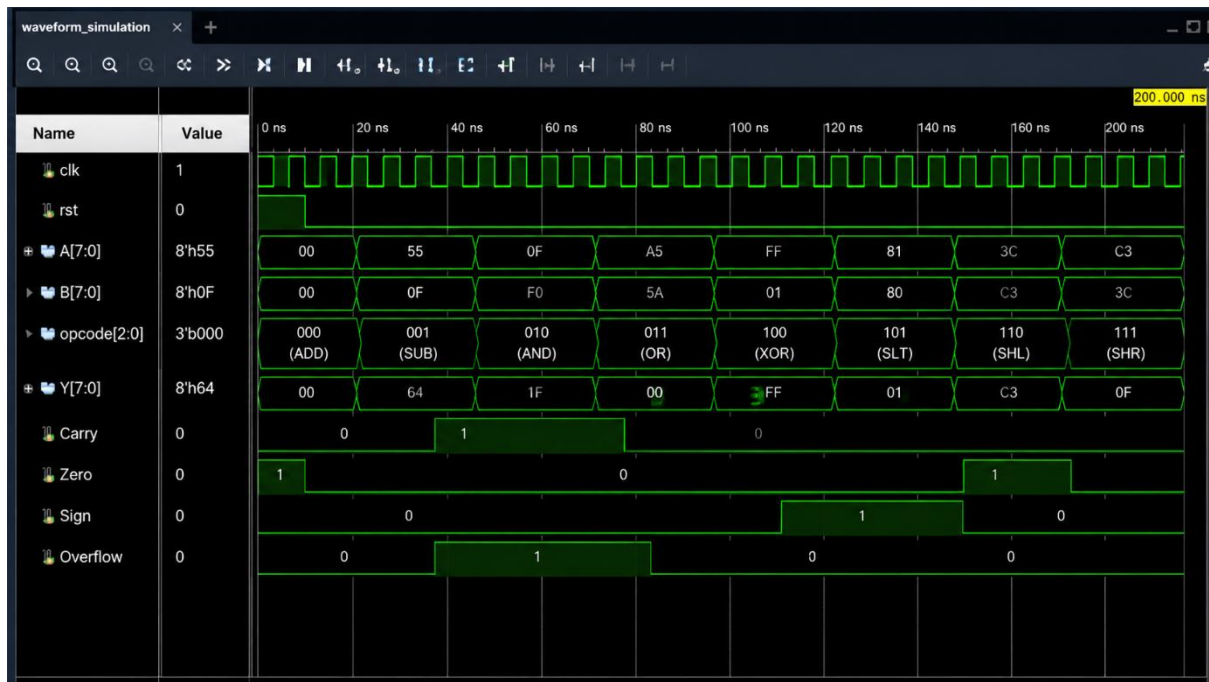


## Operations Supported:

| Opcode | Operation | Description         |
|--------|-----------|---------------------|
| 000    | ADD       | A+B                 |
| 001    | SUB       | A-B                 |
| 010    | AND       | Bitwise AND         |
| 011    | OR        | Bitwise OR          |
| 100    | XOR       | Bitwise XOR         |
| 101    | NOT       | Bitwise NOT         |
| 110    | SHL       | Logical Shift Left  |
| 111    | SHR       | Logical Shift Right |

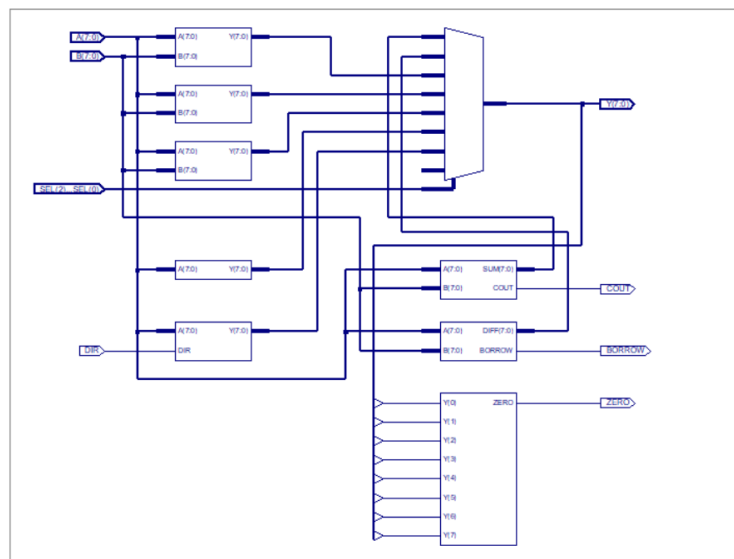
## Waveforms

The simulation waveforms generated in Xilinx Vivado were analyzed to verify the correctness of each implemented operation. The observed outputs confirmed that the ALU executed the selected arithmetic, logical, and shift operations accurately for all applied test cases.



## RTL Schematic

The RTL (Register Transfer Level) schematic provides a graphical representation of the hardware generated from the Verilog source code. It illustrates how the individual modules are interconnected to form the complete ALU before the synthesis process.



## Observations

- Arithmetic functions produced correct outputs for all tested inputs.
  - Logical operations behaved according to their expected truth tables.
  - Shift operations generated accurate left and right shifted results.
  - Simulation results matched the expected outputs, confirming correct ALU functionality.
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## **Conclusion**

The implementation of the 8-bit ALU using Verilog HDL was completed successfully. The design effectively performs arithmetic, logical, and shift operations based on the 3-bit control input. Functional verification through simulation confirmed that every operation produced the expected output. The generated RTL schematic also validated the correctness of the hardware design. This project provided practical experience in Verilog programming, digital circuit design, modular implementation, and hardware verification, making it a strong foundation for more advanced processor and VLSI design projects.